

Demystifying Bharatanatyam: A Multi-Modal Framework for Extracting Meanings and Associations in Indian Classical Dance

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Abstract

This paper introduces a comprehensive framework aimed at elucidating the meanings and associations inherent in the Indian classical dance form, Bharatanatyam, leveraging machine learning techniques and immersive technologies. This study proposes a framework to extract nuanced interpretations from body postures and sound through a multi-modal approach. There is a need for research that integrates technological advances with a deeper understanding of the psychological perception of the dance form. Therefore, a prototype was developed following the proposed framework to demonstrate its potential outcomes. Participants were invited to engage with Bharatanatyam videos both with and without labelled gestures and expressions. Results indicate that viewers comprehend the dance better when provided with gesture and expression labels. Furthermore, the framework was extended to be experienced through VR headsets, highlighting the potential for establishing a learning environment in virtual reality. This approach not only fosters cultural awareness but also facilitates the preservation and dissemination of traditional art forms, particularly among younger generations, through the integration of innovative technologies.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

Bharatanatyam, Indian classical dance, Immersive technologies, Cultural preservation

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1 Introduction

In India, the classical dance forms portray traditional culture and these dance forms involve a combination of body movements, gestures, and facial expressions to portray certain emotions and feelings in a defined stage setting. Bharatanatyam is a classical Indian dance form that is popular worldwide, yet it remains inaccessible and difficult to understand for many enthusiasts [11]. This inaccessibility is attributed to various factors such as the language, abstraction, genre of music, the difficulty in understanding the 'Pure Abhinaya'—portrayed through bodily movements and facial expressions—and the artist's practice of striving to attain the expression of spirituality through the creation of 'Bhava' and the spectator's experience of gaining 'Rasa' in the process of spiritual expression [11, 15]. *Bhava* and *Rasa* are interconnected concepts where *Bhava* is feeling and *Rasa* is the aesthetic experience derived from proper communication of emotions and situations. *Bhava* makes one experience the *Rasa*—the spiritual and aesthetic experience. This brings a lot of reluctance and resistance on the part of the performing arts enthusiasts to experience the dance form. Also, interpreting its intricate nuances poses challenges, particularly for contemporary audiences.

The preservation of cultural heritage is paramount and Bharatanatyam, with its rich history and artistic depth, is no exception. Despite its cultural significance, the dance form faces the risk of diminishing engagement, especially among younger audiences [6]. It is evident that the younger generation today is influenced by the newer mediums of entertainment and these mediums are driven by digital technology. Hence, they may find it challenging to relate to the traditional and ritualistic presentations of Bharatanatyam. This disconnect is resulting in a lack of interest towards a deeper understanding of the dance form and experiencing the creation of *Bhava* and *Rasa*. Preservation of culture has utilised different forms of artistic expression in different eras of history as shown in Figure 1 [10]. Demystify is the proposed study in this paper that attempts to digitally conserve and enhance the experience of the dance form Bharatanatyam in the modern era.



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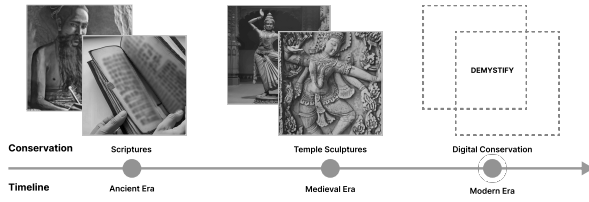


Figure 1: Historic preservation: The process of documentation, expression and meaning-making of the ancient knowledge through the generations.

This study aims to bridge the gap between traditional art and contemporary technology, presenting a framework that enhances audience comprehension and engagement through machine learning and immersive technologies. The study underscores the importance of preserving Bharatanatyam by enhancing its accessibility through multimodality.

2 Literature Review

Several studies explore the vast cultural heritage of Bharatanatyam [1, 2]. These works highlight Bharatanatyam’s storytelling aspects and the importance of the dancer’s body and expressions in conveying emotions [4, 12]. However, comprehending Bharatanatyam’s intricacies, particularly for new audiences is challenging. The integration of technology in the domain of intangible cultural heritage relating to Bharatanatyam dance has facilitated significant advancements in choreography, mentoring, performance preservation, analysis, and propagation [5, 8, 9, 13]. Past studies emphasised the presentation and documentation of Bharatanatyam dance performances using digital technologies [9, 16]. In [16], authors highlighted how Digital Dance Imaging Technology (DDIT) documents and presents ethnic dances, enhancing global accessibility while preserving cultural integrity. Another study discussed the transformative impact of augmented reality and virtual reality on Bharatanatyam dances during the Covid-19 pandemic era, highlighting the increased accessibility and introducing innovation in performance methods [5]. It also emphasised the ethical considerations and the risk of cultural misinterpretation in the process of digital dissemination.

Studies on audience perception in dance [17] and the use of technology in dance learning [13] also suggest that VR technology has the potential to enhance audience understanding and engagement. The application of machine learning and image processing in classical dance analysis has become a new prominent trend. Studies proposed an ontology-based approach for the presentation and preservation of Indian classical dance forms using the Multimedia Web Ontology Model [10]. The mentioned framework enhanced the discoverability and accessibility of dance performance videos by integrating rich contextual information. Tanwi Mallick et al. employed classifiers such as Gaussian Mixture Models (GMM) with an accuracy of 83.04% [9]. Multiple research works explored areas such as posture recognition and automatic expression recognition to understand the dance’s semantic elements [9, 20].

There is a need for research that integrates these technological advances with a deeper understanding of the psychological perception of the dance form. Addressing this gap could provide significant insights into the effectiveness of these technologies in enhancing audience comprehension and engagement. Challenges in decoding Bharatanatyam’s complex gestures and expressions necessitate innovative solutions. Multi-modal analysis of Bharatanatyam goes beyond just posture recognition to include analysis of not just facial expressions but also musical expressions, hence providing a richer understanding. Despite these extensive areas of inquiry, a notable gap remains in the literature that relates to the absence of studies testing a prototype to evaluate the psychological impact of these technological interventions on users. This study has the potential to not only enhance audience appreciation but also contribute to the preservation and dissemination of this traditional art form.

3 Method

3.1 Body Posture Extraction

This study employed image processing techniques using a pre-trained deep learning model, specifically the ResNet50, part of TensorFlow’s Keras module. The ResNet50 model was used for feature recognition, utilising a combination of Convolutional Neural Networks (CNNs) and residual connections to process and extract features from images [3]. Data for this model was extracted from publicly available videos of Bharatanatyam performances. Initially, video files were processed to remove the background. This pre-processing step ensured that only relevant visual information was used for subsequent analyses.

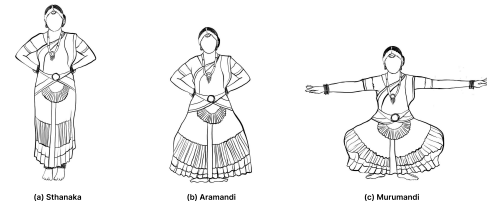


Figure 2: Bharatanatyam postures: Sthanaka, Aramandi, and Murumandi

Frames were extracted at an interval of one second from these processed videos, reducing redundancy and focusing on key visual moments involving the dancer. The frames with the background removed were resized and normalised to meet the input requirements of the ResNet50 model. Each frame was resized to dimensions of 224 by 224 pixels. Additionally, the feature vector obtained from each processed frame was characterised by having a dimension of 2048 elements. The extracted features from each frame were analysed using the K-means clustering algorithm to group similar frames into clusters based on their visual content, whose related optimisation problem is defined mathematically as:

$$\min_C \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \mathbf{c}_i\|^2$$

where k is the number of clusters, $C = \{c_1, c_2, \dots, c_k\}$ represents the set of cluster centroids, x denotes a feature vector corresponding to a frame, S_i is the set of feature vectors assigned to the i -th cluster, c_i is the centroid of the i -th cluster, and $\|x - c_i\|^2$ represents the squared Euclidean distance between feature vector x and centroid c_i .

This clustering allowed for the categorisation of the video content into segments that share visual similarities. The combination of CNN-based feature extraction and K-means clustering provided a methodology for handling and analysing video content at scale, allowing it to accomplish a detailed visual analysis, such as dance performance analysis. In this study, the clusters were analysed for three postures namely Sthanaka, Aramandi, and Murumandi, as depicted in Figure 2. The clusters were identified and labelled for each of these three postures and manually labelled as others if they did not belong to these postures. The code for the proposed model was implemented using Python [19].

3.2 Music Meaning Extraction

The audio was extracted from a video using the ffmpeg extractor which separates audio from video data. Once the audio was extracted, it underwent a process to split music and vocals using a source separation algorithm, which involved the method of spectral gating to isolate vocal tracks from background music. This isolated vocal track was further reduced for any residual noise [14]. The cleaned vocals were transcribed into text using Google’s Speech-to-Text API. Once transcribed, the text was translated into the language that was familiar to most of the subjects, in this case, English, using Google’s Translation API. Then the text was analysed for orthographic errors such as spelling variations, semantic errors that alter meanings, and syntactic errors relating to the structure of phrases.

3.3 Participants

The participants were university students, faculty and staff members aged between 25 and 54. They were familiar with English and were duly asked for their consent to participate in the experiment. They were given instructions to fill out a questionnaire after each video was shown to them. The study was conducted in an open classroom under the supervision of two researchers. Out of 13 participants, 12 participants completed all the tasks given and hence ($N = 12$) considered for further analysis.

3.4 Experimental Design

This study employed a within-subjects design where participants watched videos both without and with labels, as illustrated in Figure 3. Initially, participants viewed the performance video without any labels in Scenario 1, experiencing the dance in its unlabelled form. Subsequently, they were shown the same video with labels and conveyed the meanings of the gestures presented in Scenario 2. The videos in Scenario 2 were presented in the same order as in Scenario 1. Additionally, an interview was conducted with an expert Bharatanatyam dancer to gather insights into the dance form and to understand the nuances of the performance. This involved a detailed breakdown of the elements of Bharatanatyam to identify

and recognise the meanings conveyed by different elements and to understand the storyline derived from these elemental meanings.

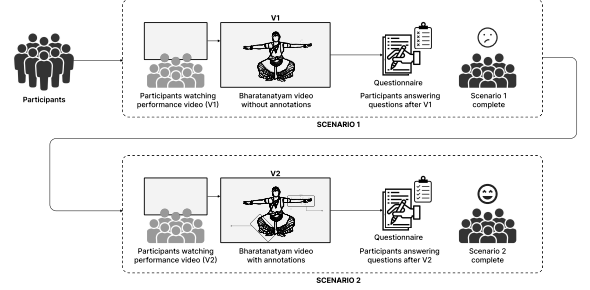


Figure 3: Experimental design: Flow diagram of within-subjects conditions

3.5 Measures

Following the sessions of the first unlabelled video, participants were asked to report their age (e) and to identify the dance form (k1), the story depicted (k2), and the meaning of the performance (k3). They were also provided feedback on their liking of the dance (a1), liking of the performance (a2) and their interest in learning about the dance form presented (a3). Correct answers to the meaning were given a score of one, while incorrect answers received a score of zero. Also, the positive liking towards the dance was given one point and the other zero. After viewing the video with labels, they were again asked to reflect on their understanding of the performance’s meaning. The labelled video prototype was developed under expert advice as a baseline for the machine learning models, which were later evaluated and compared in the results section. This approach allowed for an assessment of how labelling influences viewer’s understanding and appreciation of cultural dance performances. The overall knowledge score (K) and attitude score (A) towards the dance in the context of this study are measured using the equation below:

$$K = k_1 + k_2 + k_3 \quad (1)$$

and

$$A = a_1 + a_2 + a_3. \quad (2)$$

3.6 VR Integration

The videos were implemented as a 360-degree experience in an Oculus Rift headset. To achieve this, videos were formatted for 360-degree viewing. Next, the videos were transferred to the Oculus Rift system, ensuring compatibility with the device’s specifications. The headset was configured to a resolution of 2160 x 1200, providing a high-definition viewing experience. The refresh rate was set to 90 Hz to ensure smooth playback, and the field of view is adjusted to 110 degrees to offer an immersive visual experience. The tracking system, with 6 degrees of freedom (6 DoF), was calibrated to allow users to explore the video environment completely.

4 Results

4.1 Model Performance

The model's performance in posture classification was evaluated using a confusion matrix, from which the accuracy of the model was derived. Figure 4 depicts the confusion matrix. The matrix displays the actual postures in which the model was expected to cluster and the predicted classes containing the posture clusters.

Actual \ Predicted	Predicted			
	Murumandi	Aramandi	Sthanaka	Others
Murumandi	51	0	9	10
Aramandi	0	22	0	2
Sthanaka	0	0	12	4
Others	0	3	1	55

Figure 4: Confusion Matrix for Posture Classification

The accuracy of the model is calculated using the following equation:

$$Accuracy = \frac{\sum_i TP_i}{T} \quad (3)$$

where TP_i is the total number of true positives for posture i and T is the number of overall posture images. The accuracy of the model is approximately 82.84%. The text generated from the audio using the proposed framework was analysed for errors as mentioned in the methods. The predicted text had the errors shown in Table 1.

Table 1: Error Analysis of Audio

Error Types	Average Frequency (per sentence)
Orthographic	2
Semantic	1
Syntactic	0

4.2 Participant Data

The results for participants' proposed mean knowledge score and mean attitude score for different age groups are shown in Table 2. A paired t-test was conducted using SPSS to understand the statistical significance of the change in participants understanding of the performance. The results ($t = -2.345$, $p < 0.05$) indicate that there is a statistically significant difference between "Reported Performance Meaning - Before" and "Reported Performance Meaning - After" at the 0.05 significance level, as the p-value is less than 0.05.

Table 2: Knowledge score and Attitude score across the age groups

Age Group (years)	N	K_{mean}	A_{mean}
20-40	7	1.43	1.86
40-60	5	1.80	2.80

5 Discussion

In this study, open-source machine learning toolkits were utilised to examine the technological viability of the proposed framework. This approach allows designers to prototype faster and test machine learning-based systems efficiently. Participant responses were analysed and it was found that participants often misinterpreted the story presented to them. Out of 12 participants, only one was able to accurately report the story.

Age showed a strong correlation with a positive attitude towards learning about dance forms. This possibly indicates that the older generation has greater exposure to Indian culture and a higher appreciation for traditional art forms. The younger age group (20-40) reported a lower knowledge score ($k = 1.43$) compared to the older age group ($k = 1.80$), highlighting a cultural knowledge gap discussed in the study. However, a larger study with more participants is needed to make a conclusive and generalisable statement about the relationship between age, knowledge, and attitude.

The study highlights the importance of labelling, as demonstrated by the paired t-test results. The findings suggest that younger audiences potentially benefit more from these educational interventions. The inclusion of gesture and expression labels in the videos led to a better understanding of the dance, indicating the effectiveness of the proposed multi-modal framework. The consideration of other postures in the machine learning model allows for the expansion of the discussed model towards a broader range of postures not discussed in this study. The study witnessed that systems struggled to accurately capture and convey the nuanced meanings and context-specific references inherent in Sanskrit poetry, resulting in errors being reported. Also, the intricate linguistics of Sanskrit make it hard, especially in poetry transcriptions [18]. The postures considered in this study are three fundamental positions, and they create the foundation for developing more complex postures for future studies.

The design of this framework demonstrated a positive change in participants' understanding of the story behind the dance performance. The proposed integration of VR technology provided an immersive and interactive learning experience, potentially increasing interest and engagement in traditional art forms among younger generations. However, the proposed VR framework was not tested with all participants. The proposed approach leverages VR immersion and machine-learning-based feedback to augment the learning experience can be extended beyond traditional methods of learning [7, 21]. There were other variables not considered, such as psychological factors, randomisation, and demographic factors of the subjects who participated in the study. Additionally, the limited number of participants affected the statistical strength of our tests.

6 Conclusion and Future Works

In conclusion, this study presents a framework for decoding the complexities of Bharatanatyam through machine-learning techniques. This approach not only fosters cultural awareness but also contributes to the preservation and dissemination of traditional art forms, ensuring that their legacy stands for generations to come. The intention of this study is not to implement the experience

with always-on labels. Rather, this approach aims to aid in the initial stages by fostering awareness and enhancing motivation and curiosity to engage with such dance forms.

Future research could explore the integration of additional modalities and further investigate the long-term impact of such educational tools on cultural preservation. There is a need to address the knowledge gap influenced by modern trends and social media, which were not measured in this study. Future work could also involve user testing to evaluate the VR component's effectiveness and usability. Expanding the scope to include a larger and more diverse participant pool, as well as considering psychological and environmental factors, will provide a more comprehensive understanding of the framework's effectiveness.

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